

RANJEET KUMAR

Software Engineer | Bengaluru, India

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SUMMARY

Software Engineer at Zet (Jun 2024 – Present, intern to full-time) building fintech features on a React Native (Expo SDK) and Spring Boot stack. Shipped end-to-end UPI flows, multi-bank account management and IOB/DCB bank integrations; built custom React Native native modules in Java and Kotlin; and triage and resolve production issues using Sentry across iOS and Android. Outside work, founded SiteBot (sitebot.in), an AI support-agent SaaS with a vectorless, tool-calling RAG engine; built the self-hosted OTA, Remote Config and build-release platform that Zet adopted; and maintain Amrutkaal.com, a Panchang product computing Swiss Ephemeris astronomical data for 700,000+ cities.

SKILLS

Mobile: React Native, Expo SDK, TypeScript, React Query, React Native Native Modules (Java/Kotlin), Gradle, hot-updater OTA

Backend: Java, Spring Boot, REST API design, JPA, Node.js, Express, TypeScript, Prisma, Drizzle, Zod, Python

Data and caching: MySQL, PostgreSQL (full-text search), Redis, S3-compatible object storage (AWS S3, DigitalOcean Spaces, MinIO)

Web: Next.js, React.js, Astro, Cloudflare Pages

AI and search: LLM tool-calling agents, RAG (full-text and full-context), OpenAI API, SSE token streaming

Payments and integrations: UPI flows, bank integrations (IOB, DCB), Razorpay, Karix SMS, Mixpanel, Clerk, JumpCloud OIDC SSO

Monitoring and reliability: Sentry (@sentry/react-native), New Relic, production incident triage, staged rollouts and rollback

CI/CD and tooling: GitHub Actions, Docker Compose, Terraform, Caddy, Cloudflare Tunnel, AWS Secrets Manager, Git, Jira, Figma, Postman, Swagger

EXPERIENCE

Zet | Bengaluru, India

Fintech app: ZET UPI, digital credit cards (IOB, DCB, SBM), fixed deposits, lending. React Native (Expo SDK) + Spring Boot.

Software Engineer

Jun 2025 – Present

- Built and shipped UPI features end-to-end, including a headless Set PIN flow, a scanner shortcut/widget, lite autopay, and default payment mode selection.
- Developed the multi-banking module (bank selection UI and multibanking enhancements) so users can manage multiple bank accounts inside the app.
- Delivered IOB and DCB bank integrations covering VKYC onboarding, SIM binding skip, card activation, IOB fixed deposit payment, and IOB creation API failure monitoring.
- Implemented native Android capabilities as custom React Native native modules bridging to platform APIs for SMS, foreground services, and intent activities.
- Contributed to packages/zet-security, a custom package for native-level security protections, and maintained the Android native module codebase (13 Java and 12 Kotlin files) on Gradle.
- Structured 87 API feature modules (axios functions to React Query hooks to types) under src/api to enforce a consistent API contract layer.
- Integrated Sentry (@sentry/react-native) for real-time error tracking and crash monitoring on iOS and Android; triaged and resolved production issues spanning UPI bugs, API failures, card errors, and high-error-rate endpoint alerts, and fixed a CMS vulnerability.
- Integrated Karix SMS and Mixpanel analytics for ZET Card users to support data-driven product decisions.
- Built referral acquisition flows (referral landing page and invite screen) from Figma designs, and shipped ZET Cash post-disbursal changes plus ZET Plus UAT and 7-day free trial experiments.
- Maintained GitHub Actions workflows for automated app builds and the iOS build pipeline.

Associate Software Engineer

Dec 2024 – Jun 2025

- Contributed to UPI feature delivery, including onboarding enhancements, deferred deeplinks, and payment flow improvements.
- Contributed UI and fallback screens to the multi-banking functionality for managing multiple bank accounts in-app.
- Contributed to Spring Boot REST API design and related backend feature modules.
- Supported production reliability by triaging and resolving incidents with the team.

Software Engineer Intern

Jun 2024 – Dec 2024

- Assisted with UPI-related enhancements and mobile feature work on the React Native (Expo SDK) app, and contributed to Spring Boot REST API feature development.
- Helped integrate and validate the app monitoring and analytics components used by ZET product teams.

PROJECTS

SiteBot | sitebot.in | Founder and Developer, 2026 | Node.js, Express, TypeScript, Next.js, Clerk, PostgreSQL, Redis, OpenAI API, Docker Compose, Terraform (AWS), Caddy, Astro, Cloudflare Pages

- Founded and built an AI support-agent SaaS that crawls any website and answers visitor questions inside an embeddable chat widget; a site goes live in minutes from a URL with a single script tag, and 17,000+ pages have been answered across sites running it.
- Designed a vectorless RAG engine with two modes: small sites (up to about 24K tokens) go entirely into the prompt for zero retrieval errors; larger sites use an agentic loop where the model calls `search_site` (Postgres full-text search) and `read_page` tools until it can answer, with the first round required to search before answering.
- Made fabricated citations impossible by attaching sources on the server from the pages retrieval actually touched; capped tool rounds per plan for cost control and tracked total token usage across every model call in an answer.
- Streamed answers token-by-token over Server-Sent Events with live progress states (such as "Searching") during tool rounds, and passed the visitor's current page URL as context so answers favour the page they are on.
- Built a separate crawler service that discovers pages from sitemaps declared in robots.txt, crawls with bounded concurrency, chunks pages for full-text indexing, and re-crawls on a manual, daily or weekly schedule using content hashes so unchanged pages are never re-indexed and removed pages are cleaned up.
- Built the multi-tenant dashboard (Next.js, Clerk auth): organisations with member roles for agencies, per-website widget design stored in Postgres and served Redis-first, an embed snippet with a live preview page, daily analytics, a visitor map from geo and user-agent data, and Razorpay subscription billing.
- Ran it as five independently deployable services (API, crawler, dashboard, Postgres 16, Redis 7) under Docker Compose behind Caddy, provisioned AWS with Terraform, added backup and restore scripts and New Relic monitoring, and shipped the marketing site in Astro on Cloudflare Pages.

Zet OTA, Remote Config and Release Platform | Personal project adopted by Zet, 2026 | Node 22, TypeScript (strict), Express 5, Prisma, MySQL 8, Redis, S3-compatible storage, Next.js, React Query, Docker, GitHub Actions, OpenID Connect, Zod, pino

- Built a self-hosted over-the-air update server for the Zet React Native app (hot-updater protocol) that decides per user and per device which JS bundle to serve: the CLI pushes bundle artifacts to the API, which stores them in object storage and MySQL; every app open calls a public check endpoint and receives a bundle, nothing, or a rollback instruction.
- Layered dashboard-edited targeting rules and rollout controls on top of the library's semver, channel and cohort-based percentage selection; devices download from presigned URLs so the API never streams bundle bytes.
- Modelled the data for device and version analysis across 23 Prisma models, including per-install device state, user-to-device links, an append-only version history, OTA delivery records, a raw update-check log rolled into daily stats, daily active users and crash reports; caches request inputs rather than decisions so per-user answers stay correct.
- Built a self-hosted replacement for Firebase Remote Config: parameters, conditions and conditional values edited in the dashboard, published as immutable versions, cached in Redis and resolved per device on every fetch, with fetch statistics and an importer for existing Firebase exports.

- Added a build and release console: a developer picks platform, branch and build type, the backend records a job with a work id and dispatches GitHub Actions, the workflow uploads the artifact through a presigned PUT, and successful Android AABs are uploaded automatically to Play internal testing in version-code order behind a Redis lock; the iOS lane tracks TestFlight uploads and failures alert Slack.
- Built the Next.js dashboard (Pulse home, applications, OTA, bundles, devices, users, builds, remote config, team, audit) and replaced password login with JumpCloud OIDC SSO, role-based admin access and an audit log of every mutation; packaged as a Docker image with secrets loaded from AWS Secrets Manager.
- Diagnosed a production incident in which lowering a rollout percentage force-rolled devices back to an older bundle, traced it through production tables and the update library source, and shipped a rollout guard with regression tests.

Amrutkaal | amrutkaal.com | Next.js, Node.js, Python, Swiss Ephemeris (pyswisseph), Cloudflare, Geolocation API, i18n

- Built and maintain a free, ad-free, full-stack daily Hindu Panchang and Vedic calendar computing astronomical data from first principles.
- Serve Panchang for 546,462 Indian cities and 162,646 cities across 84 countries, each computed from local sunrise, sunset, and timezone.
- Built the astronomical engine on Swiss Ephemeris (pyswisseph) using NASA JPL DE431 with Lahiri ayanamsa, deriving Tithi, Nakshatra, Yoga, and Karana from sidereal longitudes via bisection search; accurate to ± 1 minute, with sunrise/sunset to ± 30 seconds (NOAA civil method). Validated against Drik Panchang samples.
- Implemented Kundli and matching features (Lagna, 12 houses, 9 grahas, North/South Indian and Navamsa charts, 120-year Vimshottari Dasha, Ashtakavarga, Manglik and Kaal Sarp dosha, 36-point Gun Milan) with all chart math running in the browser so user inputs are never transmitted or stored.
- Delivered daily Muhurat (Rahu Kaal, Choghadiya, Abhijit), audio Panchang, Rashifal, festival calendar (2026–2030), eclipse data, and prayer times, with English/Hindi UI, country-specific Panchang (US, UK, UAE, and others), and typo-tolerant worldwide city search.

Online Trip Management System | github.com/ranjeet-zet/Online-Trip-Management-System | Java, Spring Boot, JPA, MySQL, Postman, Swagger

- Built the backend for admin management, customer login, route/hotel/package browsing, and ticket booking, cancellation, and feedback flows; APIs documented with Swagger.

EDUCATION

B.Tech, Electrical Engineering | Punjab Technical University (PTU)

2022